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A methodology for sky-space-ground integrated remote sensing monitoring: A digital twin framework for multi-source Data-BIM integration in residential quality monitoring

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ABSTRACT

This study introduces an innovative data-BIM integrated digital twin framework for the quality monitoring residential developments. The framework amalgamates residential building information modeling (RBIM), surveying sensors and Internet of Things (SSIoT) technologies to establish a multi-faceted and multi-scale monitoring system. Within this framework, RBIM serves as a residential information twin model that aggregates comprehensive life cycle data of residential projects, encompassing design, building, operation, and maintenance. The physical things are linked to cyberspace via SSIoT technology, significantly improving data exchange and processing capabilities. To address the limitations of singular monitoring technology, a BIM collaborative multi-source data monitoring method for residential engineering quality is proposed, utilizing the BIM model alongside the advancement of real-time transmitting monitoring LiDAR, automated monitoring robots, GNSS, and various other sensors. This framework seeks to create a novel paradigm for intelligent, comprehensive data management and synthesizing heterogeneous data from several sources, focusing on the quality monitoring of residential projects to guarantee quality, safety, and operational efficiency.

1. Introduction

With the increase in the volume of the building and the complexity of the structure [1,2], the quality of residential engineering problems is becoming more and more prominent, and once an accident occurs, all of them are very likely to lead to significant loss of people's lives and property. At the present stage of China's residential engineering, quality monitoring methods generally still rely on traditional methods [3,4], mostly using a single instrument. However, now that intelligent construction has become the development trend of building construction [5–7], traditional monitoring methods have problems such as low efficiency, poor timeliness, information silos, insufficient data utilization, and high cost [8–10], which have been difficult to meet the needs of quality monitoring of residential projects. There is an urgent need to seek skill innovation to realize the intelligence of residential engineering quality monitoring, reduce building energy consumption and promote the sustainability of building materials [11,12].

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2352-7102/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Consequently, academia and industry have investigated sophisticated approaches for monitoring the quality of residential projects to address the intricate demands of contemporary residential developments [13–16]. The integrated remote sensing monitoring technique, including sky, space, and ground, utilizes various sensors through Surveying Sensors and Internet of Things (SSIoT) technology [17–19] to facilitate the real-time and complete assessment of residential project quality. Advancements in technology have elevated 3D LiDAR technology for its capacity to deliver high-precision and high-resolution 3D spatial data [20]. Compared to conventional measuring methods, Lidar technology offers rapid measurement speed, less reliance on external conditions, and the capacity for automatic processing of extensive data sets [21–23]. Nevertheless, conventional LiDAR systems sometimes need more support in their extensive applicability owing to their cumbersome dimensions, elevated expenses, and intricate operating prerequisites [24,25]. In light of this study context, exploring and developing more portable, cost-efficient, and user-friendly LiDAR monitoring systems has emerged as a significant research focus.

In order to realize the all-round monitoring of the quality of residential projects, this paper proposes a sky-space-ground integrated remote sensing monitoring methodology, constructs a data-SSIoT-RBIM integrated digital twin framework, selects multi-source sensors such as LIDAR, GPS, automatic monitoring robots and so on according to the monitoring requirements, and develops and designs a multi-sensor-integrated cloud platform for monitoring the quality of residential projects, so as to realize sky-space-ground integrated monitoring of residential projects. This framework provides a new research paradigm for intelligent big data management and multi-source heterogeneous data fusion for residential project quality monitoring.

2. Methodology

2.1. Digital twin architecture and framework

2.1.1. SSIoT-RBIM system architecture

This research introduces an SSIoT-RBIM architecture for digital twins to oversee the quality of residential construction. The architecture comprises data acquisition, transmission, digital modeling, data/model integration, and service layers (see Fig. 1).

The data acquisition layer serves as the cornerstone of the digital twin and is responsible for collecting the necessary information from various data sources. This layer deal with the challenges of heterogeneity and large amounts of data from the building level, including data of different types, formats, and sources. In particular, each sub-asset has its unique sub-data identifiers that can provide the required data, information, or models when the corresponding query is received. Various data collection technologies are used in this layer, including contactless technologies (e.g., RFID, image-based technologies), distributed sensor networks, wireless communications, and mobile access. The selection and implementation of these technologies need to be customized based on the asset type and expected data processing needs to ensure real-time data collection and efficient management.

The principal duty of the data transmission layer is to guarantee the secure and efficient transfer of gathered data to a higher level for subsequent modeling and analysis. This layer accommodates several communication technologies, encompassing short-range access networks (including WiFi, Zigbee, NFC, M2M, and Z-Wave) and broader-range communication technologies (such as 3G, 4G, LTE, 5G, and LP-WAN). As technology progresses, although WiFi remains the predominant wireless LAN technology, alternatives such as Li-Fi and LP-WAN are becoming acknowledged as viable options for creating digital twins at the city and building levels, particularly regarding security and unlicensed frequency bands. Critical concerns at the transport layer are network energy efficiency, transmission velocity, data security, and the imperative to maintain data integrity and privacy protection throughout transmission.

The data modeling layer is essential for creating and managing digital representations of physical assets. This layer utilizes Building Information Modeling (BIM) and City Information Modeling (CIM) to store intricate data regarding buildings and urban assets, incorporating external factors that influence these assets, including meteorological and socio-cultural contexts. The digital modeling layer facilitates several applications, including real-time condition monitoring, asset management, urban planning, and scenario

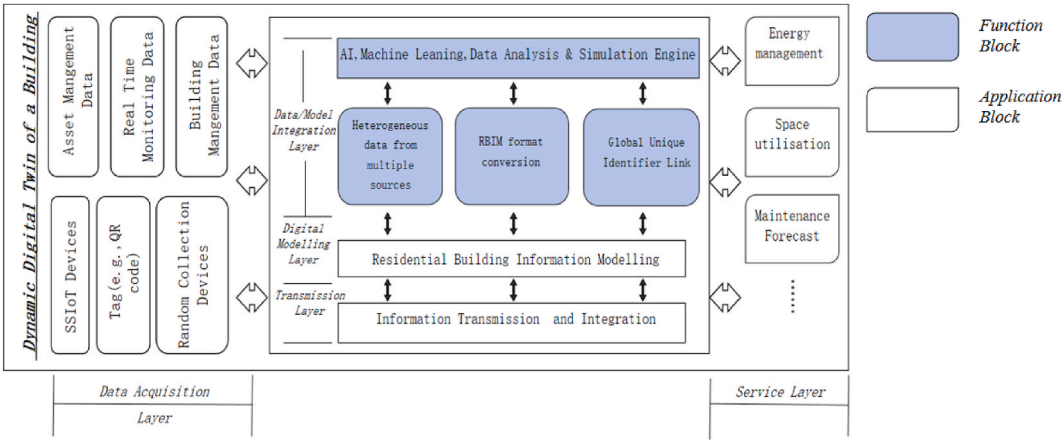


Fig. 1. SSIoT-RBIM system architecture.

simulation as a decision support system. To implement these activities, stringent models and methods must be developed to guarantee data accuracy and facilitate integration and manipulation, thereby accommodating the requirements of diverse applications, from individual facilities to entire cities.

The Data/Model Integration Layer is essential to the digital twin architecture and effectively integrates and processes all data resources. Data storage and analysis are conducted here, necessitating the implementation of advanced services such as AI-driven knowledge extraction and learning. This layer guarantees that all aggregated data and models may be updated instantaneously and respond swiftly to alterations in urban and architectural assets. Moreover, extensive data management systems and hierarchical data storage and integration technologies are essential for enhancing city-level digital twin functionality. This layer utilizes cloud computing and data visualization technologies to facilitate dynamic and efficient data administration and manipulation.

The service layer, positioned at the apex of the design, serves as the fulfillment layer that converts the insights produced by the knowledge engine (KE) and enables human contact with the data/model integration layer. It offers diverse services to the community, assesses and enhances the efficacy of the digital twin, and promotes elements such as sustainable community development, environmental management, and intelligent transportation through social feedback. This layer's functioning highlights the significance of external feedback as a knowledge input to enhance human pleasure and the overall efficacy of the digital twin via ongoing engagement.

2.1.2. SSIoT-RBIM framework

The framework is founded on the SSIOT-RBIM architecture and comprises a physical layer, a data storage layer, a data integration layer, and an application layer (see Fig. 2).

The physical layer collects raw data about the building structure and environmental conditions through multiple sensor devices deployed in the building environment. These sensors transmit the data to the upper layer for processing through a local gateway, which not only acts as a data relay but also provides initial data filtering and pre-processing. The core of the physical layer design lies in the selection and deployment of sensors. Sensors not only need to meet the requirements of high accuracy and low energy consumption but also must operate stably in the building environment over a long period. The selection of appropriate sensor types and deployment locations is fundamental to ensuring that comprehensive and accurate data can be collected. In addition, the communication protocol and security of data transmission between the sensors and the local gateway are also elements that must be considered in the design.

In the data storage layer, dynamic building data are stored within the SSIoT sensor database specially set up by the RBIM-SSIoT framework. Considering the time-varying nature of sensor data, this layer adopts an efficient time-series database to manage historical and real-time observation data and uses knowledge graphs to organize and manage static data such as attributes and structural information. By constructing relationships between entities, the knowledge graph not only stores static structural information but also combines this information with dynamic data to form a more comprehensive view of the data. Regular data backup and recovery mechanisms are implemented, as well as encryption of sensitive data, taking into account the security and integrity of the data.

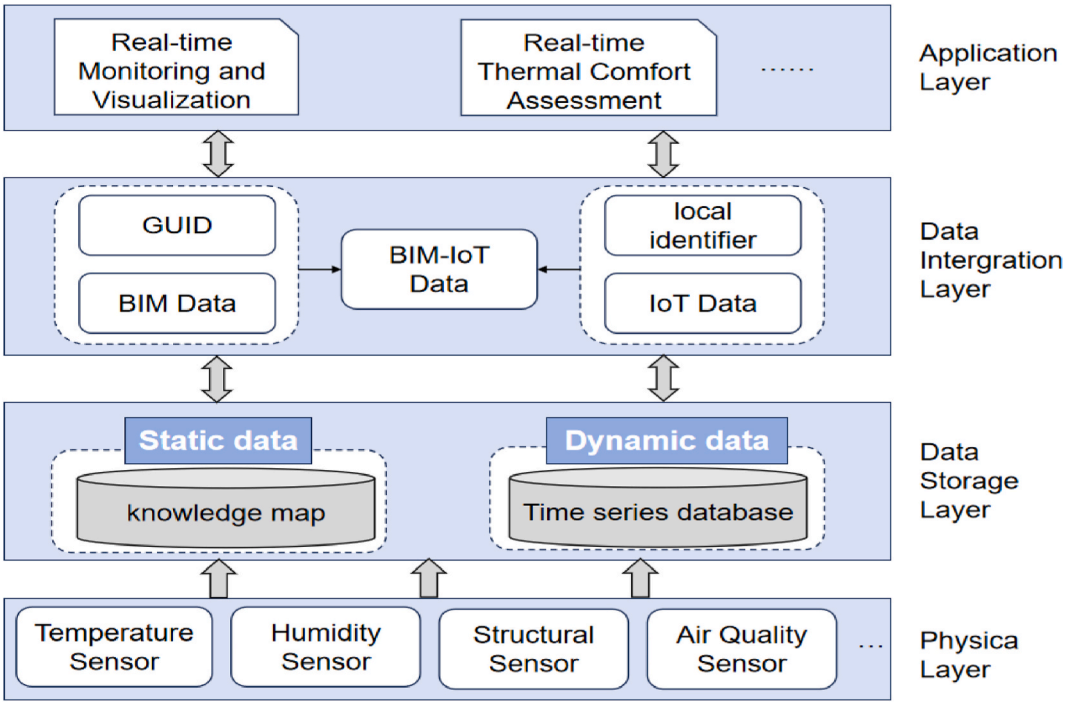


Fig. 2. SSIoT-RBIM system framework.

Residential Building Information Modeling (RBIM) data is sent in the data integration layer utilizing the Industrial Foundation Class (IFC) standard. Conversely, SSIoT data is kept in either a relational or document-oriented database. This layer facilitates the integration of RBIM and SSIoT multi-source heterogeneous data through object mapping tables. The mapping table internally correlates globally unique identifiers (GUIDs) established by sensors in the RBIM model with local identifiers for storing sensor data. This framework enables the retrieval of the associated local identifier from the mapping table by querying the sensor's GUID when SSIoT data is needed, thereby acquiring the pertinent sensor data. This integration method guarantees data consistency and traceability, establishing a dependable basis for the system's sophisticated analysis and decision assistance.

The application layer provides various application services and interfaces directly to users. In this layer, by leveraging the integrated data, various applications can be run to provide real-time data views, historical data analysis, early warning systems and maintenance recommendations to help users effectively manage and maintain building facilities. In addition, the integration of advanced data analytics and machine learning models can further enhance the intelligence of the decision support system, enabling it to anticipate building maintenance needs and potential problems based on historical and real-time data.

The established digital twin framework employs a modular data architecture in which the data from each monitoring point is treated as a separate unit. For a residential complex with hundreds of monitoring points, the modular design allows the system to perform effective data isolation and parallel processing across different monitoring points, thus reducing data processing bottlenecks. The study employs advanced data stream processing techniques to support high throughput and low latency data processing. With real-time data stream processing, the system is able to instantly receive and analyze data streams from hundreds of monitoring points, ensuring real-time and accurate data analysis in order to enhance further the scalability and flexibility of the system, a cloud-based resource management strategy is implemented, which utilizes the elastic scaling function of cloud computing to dynamically adjust resource allocation.

2.2. Real-time monitoring of multi-source sensing data

2.2.1. Automated robotic residential quality monitoring

The automated monitoring robot primarily encompasses three fundamental functions: autonomous target recognition, autonomous target tracking, and automated data recording. Automatic target identification may compute the laser beam's deviation from the prism's center and direct the operation to precisely target the prism's center utilizing the CCD array sensor's automatic motor. Automatic target tracking guarantees data continuity and precision by dynamically modifying the measuring angle and distance to monitor moving or altered points in real-time. The automated data recording feature transfers measurement data to the monitoring center or cloud platform in real time via the integrated or external data transmission module, enabling prompt analysis and long-term data storage.

The monitoring system routinely activates monitoring robots to conduct scheduled inspections of designated monitoring stations at predetermined time intervals. The monitoring stations established around the target building serve for positioning and contribute to comprehensive structural monitoring. The coordinates and elevation data of each measurement point are systematically gathered to assess the three-dimensional displacement of the residential project, which may be computed using the subsequent formula:

$$\text{Displacement} = \sqrt{(x_t\text{-}x_0)^2+(y_t\text{-}y_0)^2+(z_t\text{-}z_0)^2} \tag{1}$$

The width of the crack can be calculated by taking precise measurements of the monitoring points on both sides of the crack:

$$W = \sqrt{(x_2\text{-}x_1)^2+(y_2\text{-}y_1)^2} \tag{2}$$

Where, (x1,y1), (x2,y2) are the coordinates of the monitoring points on both sides of the crack respectively.

2.2.2. Beidou/GNSS dynamic post-processing monitoring

The mean GNSS coordinate is derived by computing the average of GNSS points on the same plane. The building orientation vector can be determined by linking the base station coordinates with the average GNSS locations, from which the building's inclination can be extracted.

The formula for the height difference ΔZ can be expressed as:

$$\Delta Z = (Z_{ev} - Z_{base}) \tag{3}$$

The horizontal distance ΔD can be calculated using the following formula:

$$\Delta D = \sqrt{(X_{ev} - X_{base})^2 - (Y_{ev} - Y_{base})^2} \tag{4}$$

The tilt (θ) is calculated as:

$$\theta = arctan\left(\frac{\Delta Z}{\Delta D}\right) \tag{5}$$

where arctan is the arctangent function, and the result is the tilt angle θ in radians. If it needs to be expressed in degrees, it can be converted from radians to degrees:

$$\theta_{degrees} = \theta \times \frac{180}{\pi} \tag{6}$$

2.2.3. SBAS-InSAR surface settlement monitoring

By establishing the time-space baseline threshold, the time-series SAR image data that satisfies the threshold criterion is segmented into many mutually independent interferometric subsets with reduced spatial and temporal baselines to ensure the coherence of the interferometric pairs. Subsequently, more precise findings for ground subsidence monitoring will be achieved.

Assuming the SAR satellite acquires N+1 SAR images at various intervals within the study area, and k differential interferograms can be derived using the specified temporal and spatial thresholds, the differential interferometric phase of the kth image in the azimuth-distance pixel coordinate system is expressed in Eq.

$$\delta\varphi_k(x,y) = \varphi(t_b,x,r) - \varphi(t_a,x,r) \approx \frac{4\pi}{\lambda}\left[d(t_b,x,r) - d\left((t_a,x,r)\right]\right. \tag{9}$$

Where: $\varphi(t_b,x,r)$、 $\varphi(t_a,x,r)$ is the phase at the moment of tb and ta; and, $d(t_b,x,r)$、 $d((t_a,x,r)$ is the cumulative amount of line-of-sight to subsidence at the moment of tb and ta. After removing the terrain residual phase, the atmospheric delay phase and various noise phases, the average rate of ground subsidence v^t can be expressed as Eq.

$$v^t = \left[v_1 = \frac{\varphi_1}{t_1 - t_0}, \ldots, v_n = \frac{\varphi_n - \varphi_{n-1}}{t_n - t_{n-1}}\right] \tag{10}$$

Then the phase is Eq.

$$\sum_{E=S_J}^{E_J}(t_k - t_{k-1})v_k = \delta\varphi_j \quad (j = 1,2,3,\ldots,n) \tag{11}$$

Where: E_J is the master image acquisition time; S_J is the slave image acquisition time; v_k is the pixel deformation rate at moment k.

From this, the matrix equation is obtained by defining $A(j,k) = t_k - t_{k-1}$ and A being a rank-loss matrix of m × n, see Eq.

$$A_v = \delta\phi \tag{12}$$

The singular value decomposition method and the least squares method can be used to find the phase value of the average subsidence rate, and then the amount of ground subsidence can be obtained.

2.2.4. Environmental sensing monitoring systems

Environmental monitoring sensors play a critical role in residential construction quality monitoring, collecting environmental data from construction sites in real-time to support project managers in assessing environmental conditions, developing construction strategies, and ensuring the safety and health of construction workers. These sensors include temperature and humidity meters, wind speed and direction meters, noise, PM2.5/10 sensors, etc., each performing a specific monitoring function. The placement strategy of the sensors is critical to ensure that effective monitoring data is obtained.

Temperature and humidity sensors should be placed away from the effects of direct sunlight and precipitation to avoid the direct effects of these environmental factors on temperature and humidity readings. At the same time, these sensors should be placed in well-ventilated areas to ensure that the temperature and humidity data measured are a true reflection of the environmental conditions on the construction site.

Wind speed and direction sensors should be placed in a manner that takes into account the ability of the building itself to block and direct the wind flow. In order to obtain accurate wind data, sensors should normally be mounted on top of the building or in an open area, above surrounding buildings and other obstacles, to minimize the effect of these obstacles on wind speed and direction.

Noise sensors should be deployed in locations representative of typical noise levels from construction activities while avoiding proximity to overly noisy mechanical equipment or major traffic thoroughfares so that excessive background noise does not affect the accuracy of the data.

PM2.5/10 sensors, on the other hand, should take into account the possible contribution of dust from wind direction and other sources. They should be placed downwind from major dust sources so that changes in air quality caused by construction activities can be accurately captured.

To mitigate the interference of environmental factors with sensor readings, research into the use of sensors with high immunity to interference and enhanced signal processing algorithms can significantly reduce data errors caused by electromagnetic and physical interference. Regular calibration and maintenance are necessary to ensure the long-term stable operation of the sensors. The system performs regular performance testing and calibration of all sensors to correct deviations that may be caused by long-term use and environmental changes.

2.3. BIM and point cloud dynamic change monitoring

2.3.1. Monitoring LIDAR measurement system integration

To augment the scanning range and versatility of the radar, we utilized a DC brushless motor to produce rotating movement, as the solid-state lidar does not possess this intrinsic capability. Integrating the DC brushless motor with the solid-state LiDAR enables the

radar to scan at a consistent angle, permitting comprehensive rotational scanning. This outcome improves the accuracy of the radar's spatial coverage and data collection. This configuration mitigates the limitations of solid-state LiDAR, hence providing improved assistance for accurate environmental modeling and target tracking. Through precise control of the speed and direction of the brushless DC motors, we can tailor the scanning method to fulfill the particular requirements of the application, resulting in improved efficiency and versatility in spatial detection. This innovative integration offers exceptional prospects for navigation and positioning in complex environments, ecological awareness, and other advanced applications.

Independent research and development of data processing procedures through the standard matrix division and secondary filter processing to achieve efficient extraction of the target ball, using a five-point quadratic function smoothing filter and RANSAC algorithm for the ball center coordinate solution, to achieve the three-dimensional dynamic real-time monitoring of building safety. As shown in the Fig. 3, the equipment breaks through the real-time transmission, point cloud data processing, the target ball center coordinate solving, and other technical difficulties, the cost of LiDAR monitoring equipment from the original tens of thousands of yuan to within 10,000 yuan, significantly reducing the price, the volume from 8 kg to 2 kg, the monitoring accuracy from 2 cm to 5 mm, real-time capture the displacement changes of the target ball, to obtain the displacement changes of the critical nodes of the building, so that the LIDAR monitoring equipment is more suitable for a wide range of application scenarios, and promotes the development of real-time LIDAR monitoring equipment in the direction of universalization.

2.3.2. Comparison of BIM and point cloud analysis methods

The dynamic change detection system utilizing BIM and point cloud data is a crucial novel method for monitoring building safety, with detection outcomes predominantly conveyed through reports. Real-time online dynamic comparison and analysis of BIM and point cloud data is a technical difficulty. A method for organizing data within the BIM model and point cloud is provided, employing a 'three-step' strategy of digital-module decoupling, geometric conversion, and attribute mapping to address issues of inadequate data interoperability and integration. This study investigates the dynamic change detection approach for BIM models and point cloud data to address the challenge of comparing and evaluating BIM and point cloud data in three-dimensional space, facilitating the efficient integration of both data types (see Fig. 4).

A dynamic change detection analysis and visualization method combining BIM and point cloud data is proposed to obtain 3D deformation results of buildings in real time. Distinguishing from the existing research, a BIM and point cloud data organization method is proposed to construct a 3DTiles tile structure of point cloud and BIM models to achieve effective integration of BIM and point cloud data. The IFC file is separated and reconstructed by component level through digital-modular decoupling to achieve error-free linking of geometric and semantic information of components; for the storage and processing problems of large-scale point cloud data, an octree segmentation algorithm is adopted. The octree refines the field point cloud data into a multilevel hash file by recursively partitioning the three-dimensional space. This data structure effectively reduces the redundancy of data storage and greatly improves the efficiency during spatial queries. The octree is able to partition the point cloud data into multiple small spatial units, which can be loaded and processed as needed. Through the spatial division of point cloud data, the system is able to quickly locate part of the data related to the BIM model, thus greatly improving the efficiency of comparison and analysis between point cloud data and the BIM model. The use of Octree not only improves the speed of data loading and rendering but also effectively optimizes memory management, especially when facing large building models containing millions of point cloud data, and still maintains high performance.

We study the iterative nearest-neighbor algorithm to find the nearest mesh model of the 3D point cloud and construct an octree and mesh cross-list to quickly identify the differences between the BIM model and the actual point cloud data so as to solve the problem of

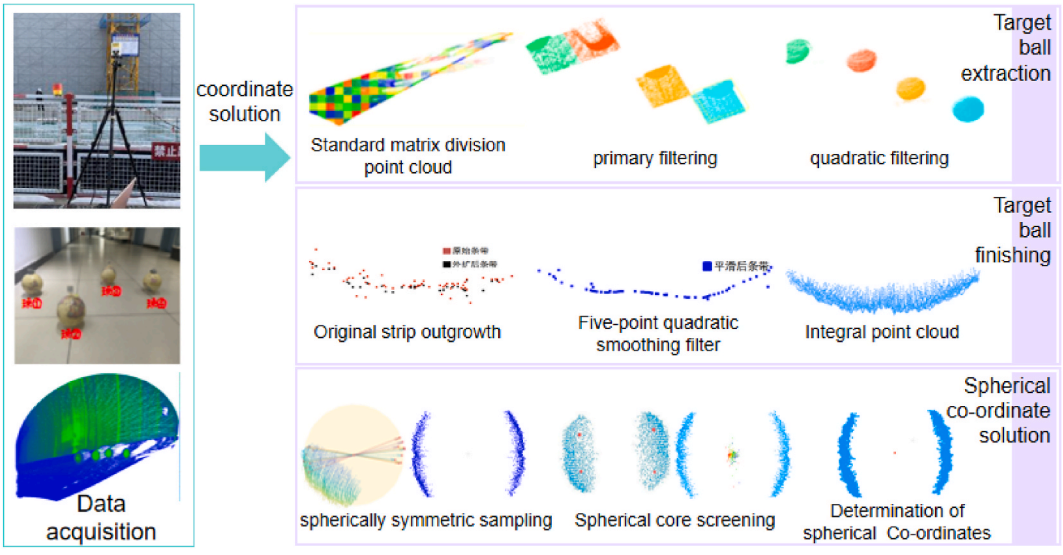


Fig. 3. Monitoring LIDAR measurement system.

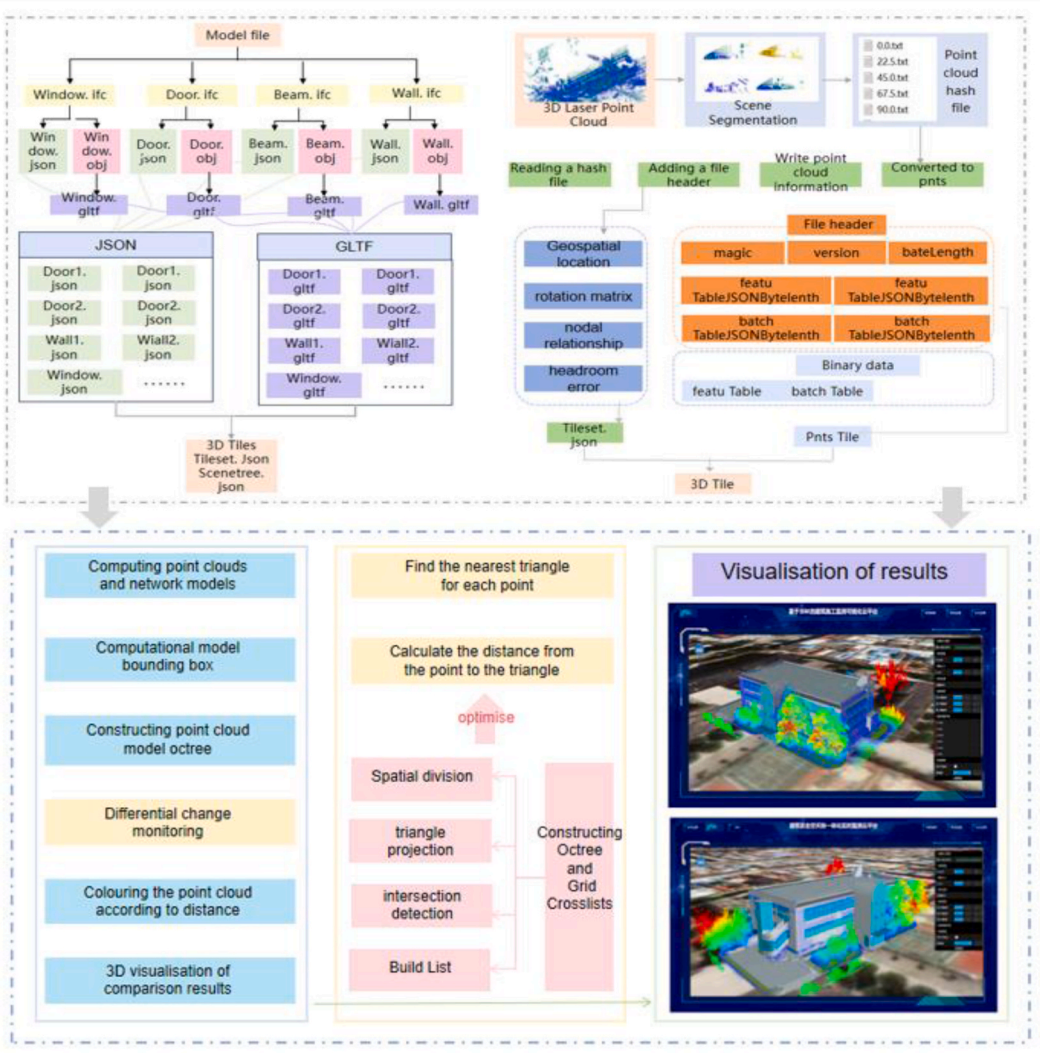


Fig. 4. BIM with point cloud dynamic change monitoring.



Fig. 5. Framework building scene.

comparing and analyzing the BIM and point cloud data in the 3D space. When dealing with large-scale datasets, the traditional single-computer computing model is often difficult to meet the real-time and high-efficiency requirements. This study adopts parallel computing and distributed processing techniques to improve computational efficiency further. By dividing the tasks of point cloud data processing and BIM data analysis into multiple subtasks and assigning these tasks to multiple computing nodes for parallel computing, the computational load can be effectively shared to avoid the bottlenecks that may occur during single-node processing. Especially in the case of large-scale data sets and extremely high density of point clouds, distributed computing can significantly shorten the computation time under the premise of ensuring the analysis accuracy. This optimization not only improves the data processing efficiency but also provides strong technical support for real-time dynamic change detection and visualization.

3. Experiments and analyses

3.1. Multi-source remote sensing monitoring experiments

3.1.1. Monitoring of frame buildings

A monitoring LIDAR measurement system was employed to observe a frame construction (see Fig. 5). Two distinct target spheres were positioned at varying elevations within the building for an eight-phase point cloud data acquisition, and the alterations in the centers of the individual table target spheres were examined to assess the modifications in the building.

The point cloud data obtained from the acquisition were processed and analyzed (see Fig. 6). The particular target ball was extracted (see Fig. 7), and the data were processed to calculate the spherical center coordinates using the scanline-expanded local interpolation smoothing algorithm (LSIIS). The coordinates of the two unique target balls for the eight phases are shown in Table 1.

This study employs a comprehensive analytical approach to illustrate the building's deformation trend over time precisely. Utilizing the coordinate data of the marker sphere collected during the initial period as a reference baseline, we monitor the displacement variations of the sphere's center for each marker sphere across the x, y, and z axes in successive periods (refer to Table 2). By analyzing the coordinate offsets of each period relative to the reference period, we can meticulously measure the motion and deformation of the building structure in all directions. Furthermore, the axial change data for each marker sphere is represented as a time series of line graphs (see Fig. 8), enhancing the intuitiveness of the results and aiding in the detection and analysis of change trends.

Through the examination of the positional alterations of the two primary target spheres during the eight phases of data collection, we discern the following phenomena: The positional variation of target sphere ① remains below 1 cm, indicating a degree of stability, but the positional variation of target sphere ② extends to within 2 cm, demonstrating comparatively less stability. Target ball ① is situated at the base of the building, and its minimal positional variation signifies the structural stability of the foundation; conversely, target ball ②, positioned at the apex of the building, exhibits more pronounced fluctuations, suggesting a more extraordinary displacement phenomenon in the upper section of the structure. Subsequent investigation indicates that the fundamental cause of the disparity in displacement between the upper and lower sections is the instability of the beam connections in the steel frame structure.

This discovery is crucial for evaluating the structural integrity and stability of edifices. In steel-framed structures, the stability of beam connections is directly linked to the total structural integrity. This study elucidates the disparities in building stability across varying heights. It underscores the necessity of enhancing the resilience of structural joints, offering critical insights for future maintenance and reinforcement efforts.

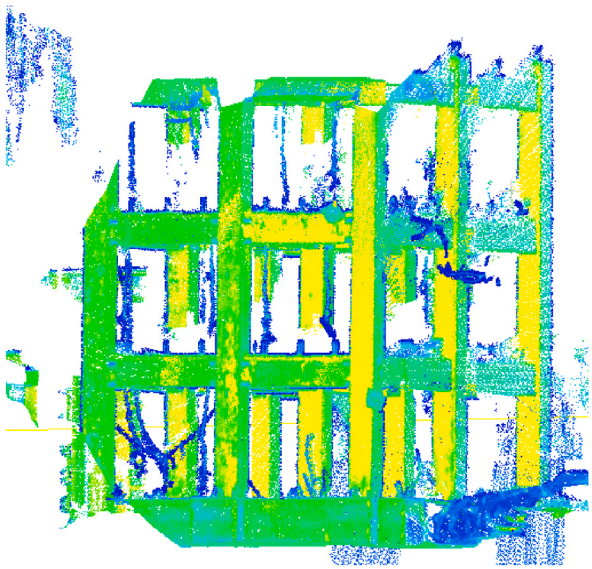


Fig. 6. Frame building point cloud data.

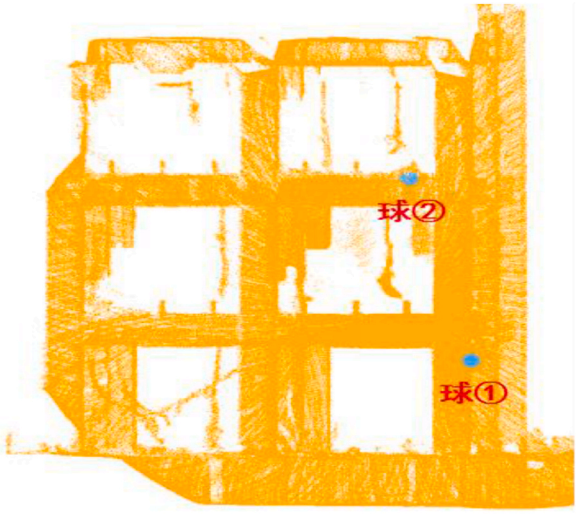


Fig. 7. Target ball extraction.

Table 1
Spherical coordinates of different periods.

	Type	first phase	second phase	third phase	fourth phase	fifth phase	sixth phase	seventh phase	Eighth phase
Sphere 1	X	−245.597	−245.596	−245.593	−245.590	−245.591	−245.588	−245.591	−245.593
	Y	174.203	174.206	174.209	174.206	174.201	174.205	174.203	174.200
	Z	68.9991	68.994	69.007	68.992	68.996	69.001	68.991	68.994
Sphere 2	X	−245.612	−245.614	−245.600	−245.607	−245.594	−245.590	−245.599	−245.602
	Y	174.651	174.666	174.674	174.672	174.659	174.650	174.663	174.664
	Z	70.955	70.954	70.963	70.965	70.951	70.936	70.940	70.951

Table 2
The amount of change in coordinates.

	Type	second phase	third phase	fourth phase	fifth phase	sixth phase	seventh phase	Eighth phase
Sphere 1	ΔX	0.001	0.004	0.007	0.006	0.009	0.006	0.004
	ΔY	0.003	0.006	0.003	−0.002	0.002	0	−0.003
	ΔZ	−0.0051	0.0079	−0.0071	−0.0031	0.0019	−0.0081	−0.0051
Sphere 2	ΔX	−0.002	0.012	−0.005	0.018	0.022	0.013	0.01
	ΔY	0.015	0.023	0.021	0.008	−0.001	0.012	0.013
	ΔZ	−0.001	0.008	0.01	−0.004	−0.019	−0.015	−0.004

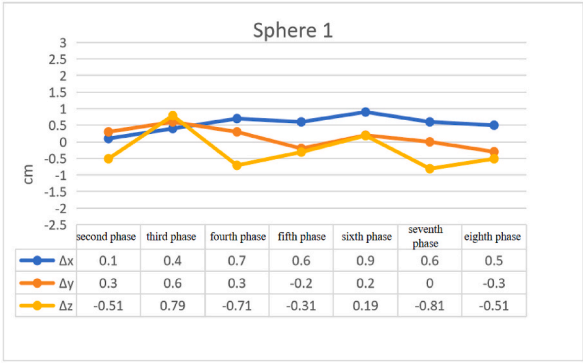
3.1.2. Tower crane monitoring

This study developed a dynamic scanning protocol for ongoing data acquisition during the operation of a tower crane at a building site (see Fig. 9). The objective was to document the crane's detailed condition at various intervals.

Two sets of 100,000-point cloud data were obtained at distinct scanning intervals (see Fig. 10). The tower crane's geographical position and structural modifications at these successive time intervals were meticulously analyzed and contrasted utilizing advanced data techniques, thereby precisely illustrating its operational dynamics.

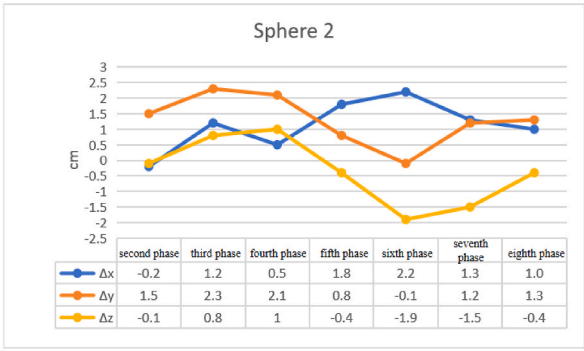
The results of the comparative analyses revealed that the average offset of the tower crane during operation exhibited a positive displacement of 0.77 cm and a negative displacement of −0.78 cm (see Fig. 11). This result implies that during the monitoring period, the trajectory of the tower crane moved forward in the north direction by an average of about 0.77 cm and backward in the other direction by an average of about 0.78 cm. This slight bi-directional offset reveals the stability characteristics of the tower crane during operation and, at the same time, hints at the possible existence of cyclic oscillations or structural offsetting behaviors.

Further analyses suggest that the determination of this offset not only reflects the elastic response of the crane structure during stresses but may also indicate the effects on the crane positional accuracy due to factors such as wind loads, load variations, and operational practices. In addition, this data is of great value in assessing the operational safety and structural health of the tower crane, providing a quantitative means to monitor the real-time operational condition of the crane and providing data support for taking necessary maintenance or adjustment measures.



	second phase	third phase	fourth phase	fifth phase	sixth phase	seventh phase	eighth phase
Δx	0.1	0.4	0.7	0.6	0.9	0.6	0.5
Δy	0.3	0.6	0.3	-0.2	0.2	0	-0.3
Δz	-0.51	0.79	-0.71	-0.31	0.19	-0.81	-0.51

a)Sphere 1 Coordinate Change



	second phase	third phase	fourth phase	fifth phase	sixth phase	seventh phase	eighth phase
Δx	-0.2	1.2	0.5	1.8	2.2	1.3	1.0
Δy	1.5	2.3	2.1	0.8	-0.1	1.2	1.3
Δz	-0.1	0.8	1	-0.4	-1.9	-1.5	-0.4

b)Sphere 2 Coordinate Change

Fig. 8. The amount of change in the center of the ball.

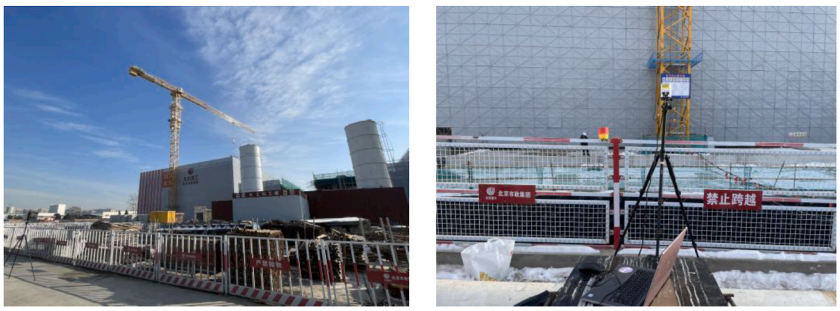


Fig. 9. Tower crane scene.

3.1.3. Accuracy analysis

An experimental comparison is carried out to verify the accuracy of the 'Scanning Line Expansion Local Interpolation Smoothing Algorithm' in the centering of the custom-made target ball. The portable LiDAR Livox, total station, and Faro were used in the experiment to collect data from the particular target sphere (see Fig. 12).

For the diagonal reflectors of each target sphere, measurements were made using a total station to obtain the center coordinates of the two reflectors, and the center was solved to obtain the sphere center coordinates, which were considered the actual values. Subsequently, the coordinates were solved for the data collected by Livox using a self-developed scanline-expanded local interpolation smoothing algorithm (LSIIS), as well as the commonly used Weighted Total Least Squares (WTLS) and Iterative Closest Point (ICP) algorithms. WTLS and ICP were used to calculate the spherical center coordinates for the Faro data. LSIIS is not used for the Faro data because its data quality is highly accurate, does not fluctuate, and does not require quality optimization processing. The spherical center coordinates were obtained by different instruments using different methods, as shown in Table 3. Since the coordinates of the total station, portable LiDAR, and faro data are not under the same coordinate system and cannot be compared directly, the spherical center distance of different target balls is used to compare the accuracy performance of the assessment algorithm, and the specific

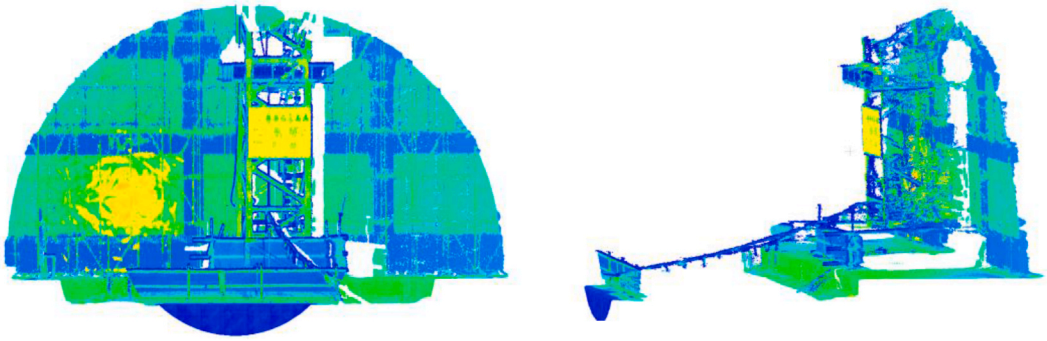


Fig. 10. Tower crane point cloud data.

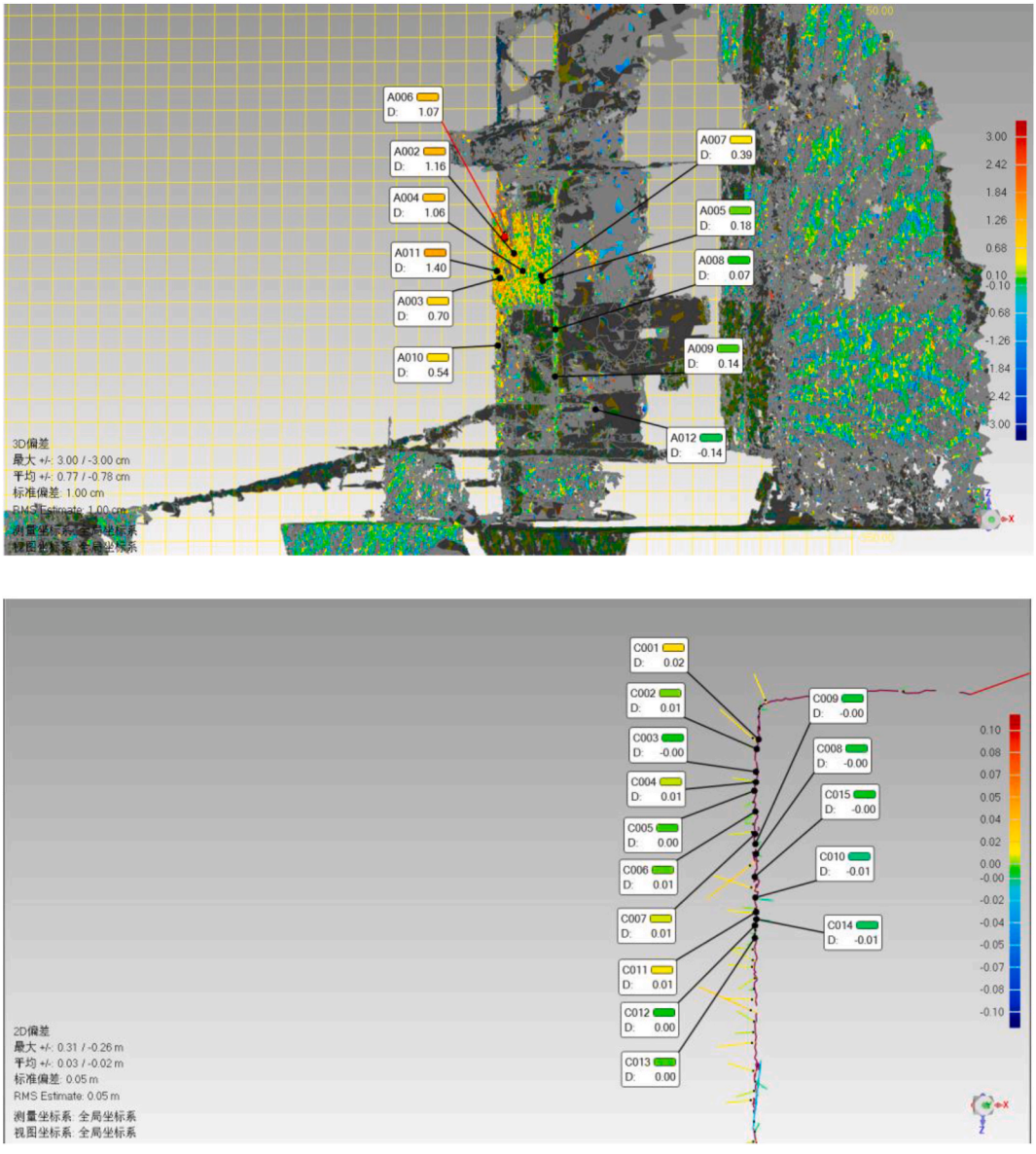


Fig. 11. Tower crane data analysis.

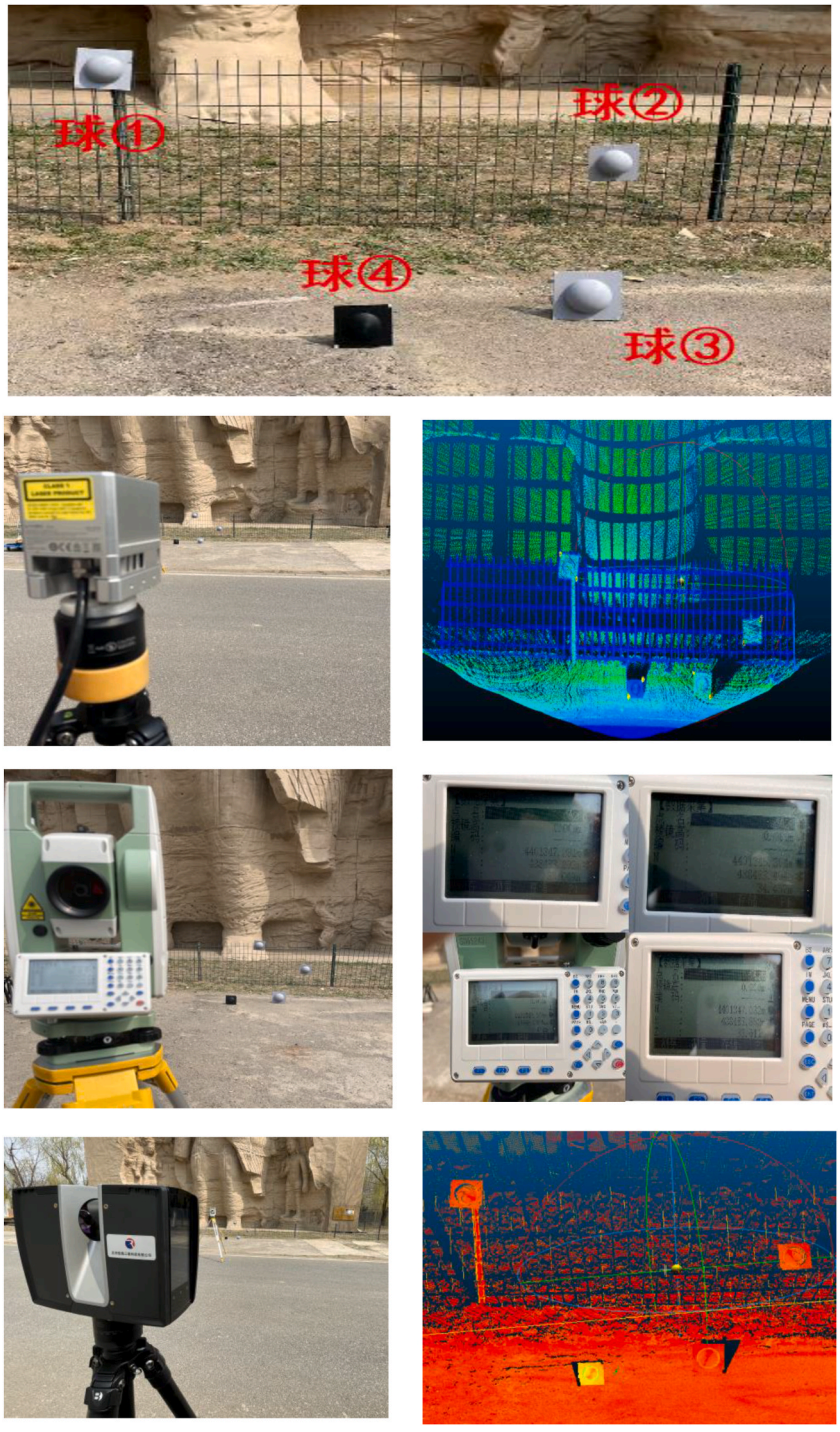


Fig. 12. Accuracy verification.

calculation results are shown in Table 4.

In the verification experiments of the accuracy of the particular target ball center acquisition, the data collected by the same instrument with different algorithms were used to calculate the center coordinates of the ball. The comparison results were shown more intuitively through the bar charts (see Fig. 13), in which the yellow bars represent the center error of the target ball data collected by the portable LIDAR system (Livox) with the self-developed LSIIS algorithm.

The graphics indicate that the Faro LiDAR data is processed with the highest accuracy by both the Point Cloud Library (PCL) and

Table 3
Sphere center coordinates.

Total station									
		x			y			z	
Sphere1		4401345.3785			438483.678			34.375	
Sphere2		4401346.747			438482.040			33.881	
Sphere3		4401347.309			438483.2435			33.56	
Sphere4		4401347.0065			438483.9135			33.5425	
Monitoring LiDAR									
	LSIIS			WTLS			ICP		
	x	y	z	x	y	z	x	y	z
Sphere1	5.4534	0.53454	0.54493	5.4748	0.5331	0.55199	5.4704	0.53344	0.5450
Sphere2	5.3030	−1.44016	0.09834	5.3198	−1.445	0.09844	5.3220	−1.4417	0.1014
Sphere3	4.0985	−0.17398	−0.2186	4.1273	−0.889	−0.2189	4.0983	−0.8806	−0.217
Sphere4	3.8978	−0.16646	−0.2518	3.8788	−0.164	−0.2488	3.9185	−0.1634	−0.255
Faro									
	WTLS			ICP					
	x	y	z	x	y	z			
Sphere1	346.685	145.259	52.6438	346.685	145.259	52.6439			
Sphere2	346.221	143.347	52.1492	346.22	143.348	52.1492			
Sphere3	345.125	144.1	51.8272	345.125	144.1	51.8278			
Sphere4	345.055	144.837	51.809	345.055	144.837	51.8094			

Table 4
Sphere center distance and center error calculation.

	1–2	1–3	1–4	2–3	2–4	3–4
Total station	2.191	2.139	1.845	1.366	1.921	0.735
Livox-LSIIS	2.1957	2.1372	1.8412	1.3691	1.9245	0.7325
difference	−0.0027	0.0028	0.0028	−0.0031	−0.0035	0.0025
inaccuracy	0.004194m					
Livox-WTLS	2.035	2.105	1.917	1.354	1.959	0.766
difference	0.156	0.034	−0.072	0.012	−0.038	−0.031
inaccuracy	0.0744m					
Livox-pcl	2.0298	2.1127	1.8796	1.3835	1.9315	0.7404
difference	0.1612	0.0263	−0.0346	−0.0175	−0.0105	−0.0054
inaccuracy	0.0268m					
Faro-WTLS	2.03	2.11	1.88	1.37	1.92	0.74
difference	0.0008	0.002	−0.001	−0.004	0.001	−0.001
inaccuracy	0.00205m					
Faro-pcl	2.1887	2.137	1.846	1.3683	1.92	0.736
difference	0.0023	0.0020	−0.0010	−0.0023	0.0010	−0.0010
inaccuracy	0.0017					

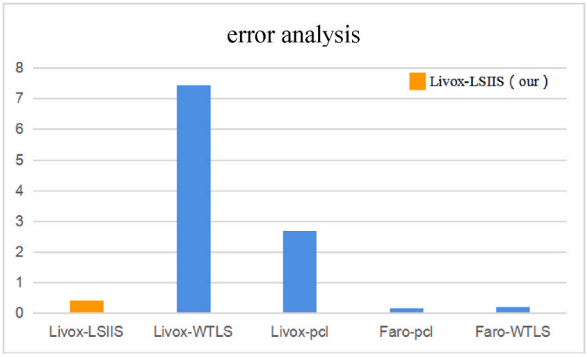


Fig. 13. Error analysis.

WTLS algorithms. The fundamental cause of this phenomenon is that the Faro data possesses exceptionally high raw accuracy, achieving precision at the micron level. Consequently, when these methods are employed to compute the spherical center coordinates, the precision of the spherical center coordinates derived from the Faro data is inherently superior to that of the self-developed portable LiDAR monitoring system.

Utilizing Livox LiDAR for data collecting and employing the LSIIS for processing yields a median error of 0.004194 m, representing the most fantastic accuracy among all processing methods utilizing Livox data. In contrast, the Livox data yielded a median inaccuracy of 0.0744 m when analyzed using WTLS and 0.0268 m when evaluated with the PCL. The LSIIS algorithm demonstrates superior accuracy in determining the central position of the specifically marked target ball when analyzing Livox lidar data, significantly surpassing the performance of the WTLS and PCL algorithms. This indicates that although the Faro laser scanner has exceptional accuracy, the Livox Lidar can offer a comparably superior alternative when paired with the LSIIS algorithm.

3.2. *Digital twins for real-time monitoring of multi-source sensor data*

This study takes the cloud platform as the core and relies on the cloud server to construct a cloud database for receiving, processing, storing and forwarding multi-source data generated in residential engineering quality monitoring, such as environmental monitoring data, building structural safety monitoring data, and point cloud data. Using the elasticity and scalability of the resources of the cloud platform, a digital twin system based on RBIM for monitoring the quality of residential works is developed and designed with the server as the support and the SSIoT system architecture. The system is able to call the information table and data table stored in the cloud database through the back-end service to achieve dynamic management and scheduling of data. The end devices in the field are configured with a local server and an edge control unit designed to collect data from the end sensors while storing this data and forwarding it to the cloud database. In order to ensure that data are not lost or corrupted during transmission between the local server and the cloud database, this system is designed with data integrity and security safeguards. Firstly, to prevent loss or damage during data transmission, the system adopts a distributed data backup mechanism. The local server caches sensor data in real time and synchronizes it with the cloud database during network recovery. Each data transmission is accompanied by an integrity check to ensure that the data is not damaged during transmission. If the data is corrupted during transmission, the local server will automatically re-transmit the data through the retransmission mechanism until the cloud database successfully receives and verifies the data. Secondly, in order to improve the fault tolerance of the system, the local database will be used as a backup source for the cloud database in the event of network interruption or security attack on the cloud server. This backup function not only ensures high data availability but also ensures data security through encrypted storage. To prevent data loss or tampering, the data synchronization between the local database and the cloud database adopts a two-way confirmation mechanism, i.e., before and after each data synchronization, the system will carry out a data consistency check to ensure the consistency of the data during synchronization through timestamp control. The local server and the edge control node are connected through various communication modules (e.g., WiFi, Bluetooth) and interfaces based on the Windows operating system to efficiently summarise and process data from various types of sensors (see Fig. 14).

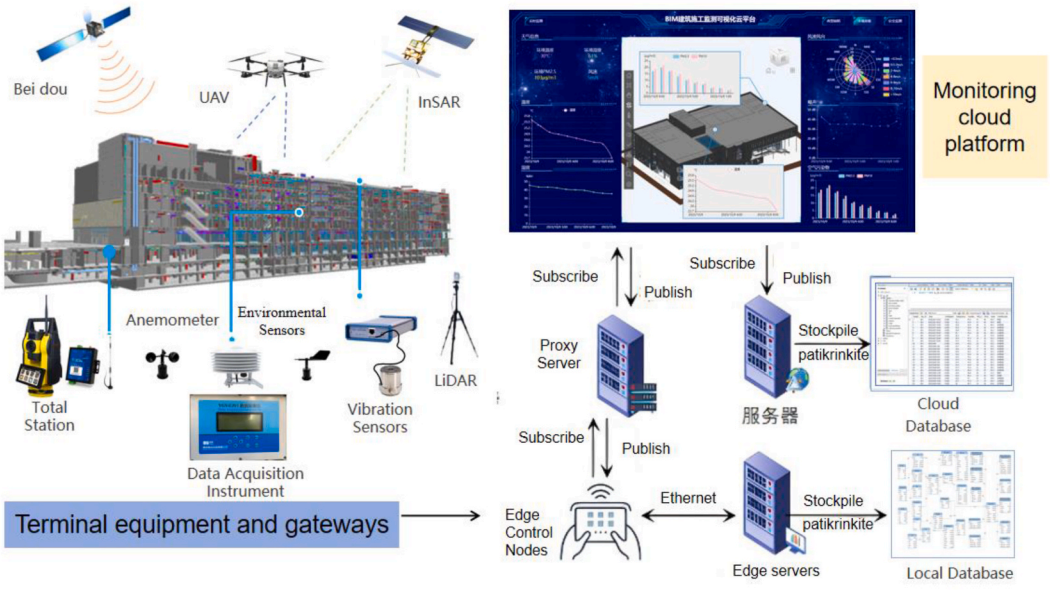


Fig. 14. Cloud platform for real-time monitoring of multi-source sensing data.

3.2.1. Design of a spatial database for integrated remote sensing of space-air-ground

An object-relational spatial database is proposed to address the challenges of efficient storage and management of multi-source monitoring data in long-term monitoring activities. The conceptual and logical models of the space-air-ground integrated remote sensing monitoring spatial database are designed, and the model's entities, attributes, and connections are discussed. In this paper, following the standardized design methodology and targeting the critical needs of quality monitoring of residential construction projects, a spatial database management system with the functions of data storage, management, analysis, and visualization is established based on the conceptual and logical models by adopting a mainstream backend framework and combining the development with common database and programming language tools as well as BIM technology. The system can process monitoring data efficiently and stably, has the advantages of fast response and low error rate, effectively improves the storage and management efficiency of space-air-ground integrated remote sensing monitoring data, and achieves comprehensive analysis and management of sky-space-ground integrated remote sensing monitoring data (see Fig. 15).

3.2.2. Analyses of integrated remote sensing monitoring by space-air-ground

This work established a generic family model to efficiently describe sensor devices within the RBIM, encapsulating all pertinent sensor devices. The sensor families in this model are positioned appropriately within the BIM model to represent their precise and accurate placements in the physical structure. To systematically manage the information of these devices, two-parameter groups are established for the sensor families, categorized as 'Identity Data' and 'Building Attributes.' Constructing Properties. Identity Data refers to the section of the Type Properties box that retains identifying parameters, like the sensor's name and position. The Building Properties section in Revit's Type Properties window primarily documents the environmental parameters recorded by the sensor. The sensors included in the RBIM model establish GUID that facilitate the precise and unique mapping of the sensors. Fig. 16 illustrates the integration of sensor data with the RBIM model, enabling the RBIM model to serve as the data's conduit, creating a digital twin of the physical and virtual realms of engineering.

Contemporary home engineering monitoring initiatives depend on integrating diverse data from many sources for precise spatial positioning and structural stability assessment. Automated monitoring robots deliver precise coordinate information and altitude measurements. Simultaneously, GNSS is tasked with delivering worldwide positioning services. SBAS-InSAR technology observes the subsidence of building structures by examining the phase discrepancies between radar images captured at various intervals. The data integration layer of the SSIoT-RBIM system effectively integrates these multi-source data. The time-series data collected by the automated monitoring robot is utilized to compute building settlement, and the resultant settlement data is compared and merged with the time-series settlements obtained by the SBAS-InSAR technique. In the data integration process, settlement data are linked to GUID specified in the BIM via an object mapping table, ensuring that data from each measurement point accurately corresponds to the precise location of the physical structure. Fig. 17 illustrates that the settlement monitoring outcomes from the autonomous monitoring robot and the SBAS-InSAR technique at the same monitoring location exhibit identical settlement patterns, achieving millimeter and sub-millimeter precision. This high-precision monitoring dataset offers robust scientific support for project management and decision-making, improving the capacity to anticipate and avert structural issues.

To ensure data consistency and deal with potential data conflicts, the system uses a weighted average method to integrate

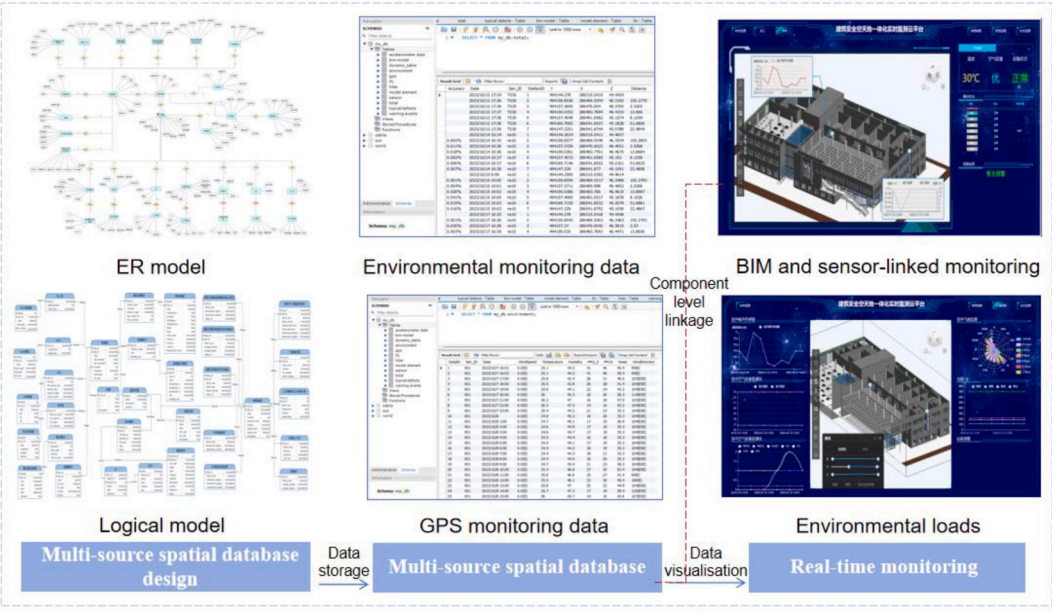


Fig. 15. Integrated space-air-ground remote sensing monitoring database.

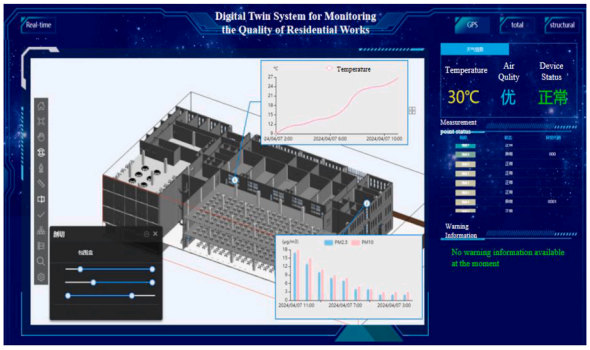


Fig. 16. BIM and sensor linkage.

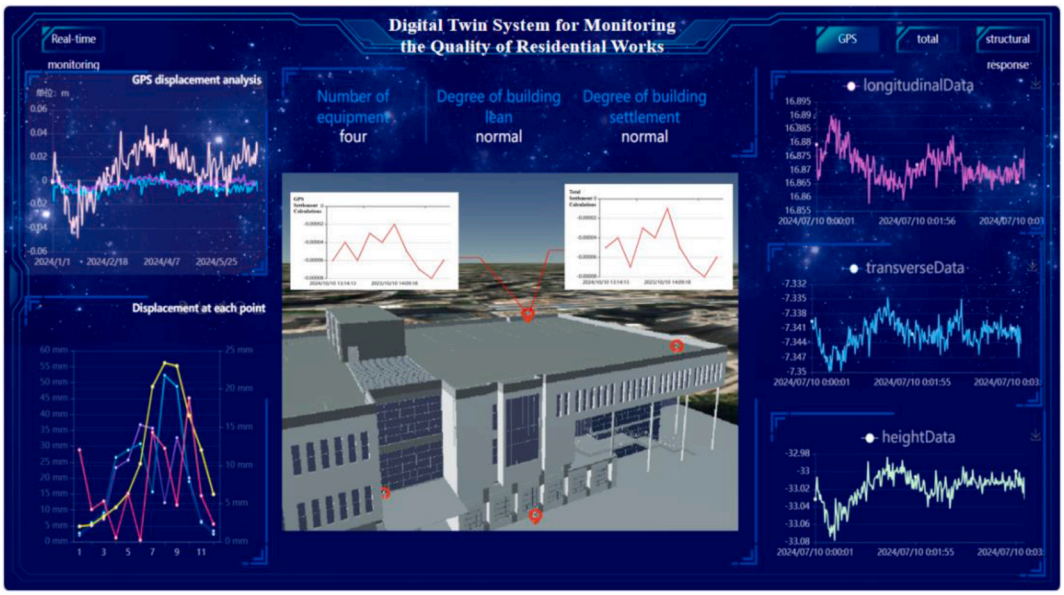


Fig. 17. Building settlement monitoring analysis.

measurements from various data sources, optimizing the overall data accuracy by assigning appropriate weights (based on their historical accuracy and reliability records) to data from different sensors. The system also features a conflict detection mechanism for identifying and handling discrepancies from different data sources in real-time. This mechanism identifies discrepancies that exceed a predefined threshold by analyzing time-series data generated by different sensors at the same monitoring point for comparison. Upon detection of a potential data conflict, the system automatically initiates a resolution process, which may include but is not limited to recalibration of sensors, re-collection of data, or correction of data suspected of being incorrect using statistical methods.

4. Conclusion

(1) This paper proposes a digital twin framework based on sky-space-ground integrated remote sensing monitoring, which is capable of automatically integrating multi-source sensing data and achieving unified data processing and storage through a cloud platform. It not only improves the efficiency of data processing but also avoids the problem of information islands existing in the existing system, thus providing a new paradigm of data fusion and information integration for the operation and maintenance management of residential projects. Through automated and intelligent monitoring, this framework significantly reduces the investment of manpower and material resources and drastically lowers the maintenance cost of the project.

(2) A multi-sensor-integrated digital twin system for monitoring the quality of residential projects is developed and designed relying on the SSIoT-RBIM system architecture. In practice, the SSIoT-RBIM framework involves the integration of a large number of data from different sources and formats, especially when dealing with heterogeneous data from multiple sources; there are problems with standardization and data format conversion. The BIM model and the point cloud data come from different devices and platforms, while the point cloud data may come from a laser scanner, a drone, or other sensors with

various data formats. How to ensure that these different formats of data are effectively integrated into a unified framework is a major challenge in realizing the SSIoT-RBIM system.

(3) The current system is limited to the integration and visual presentation of data from multiple sources, and future research needs to be further extended in terms of prediction and assessment by integrating advanced data analytics and machine learning models, making it possible to predict the maintenance needs and potential problems of a building based on historical and real-time data.

CRediT authorship contribution statement

Ming Guo: Supervision, Conceptualization. **Xiangling Wu:** Writing – original draft. **Huihui Qi:** Methodology. **Ying Zhang:** Data curation. **Jiangjihong Chen:** Software. **Yaxuan Wei:** Software. **Xiaolan Zhang:** Visualization. **Xiaoke Shang:** Data curation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] H. Yang, S. Jiang, Stress point monitoring algorithm for structure of steel cylinder concrete pipes in large buildings, Symmetry 14 (6) (2022) 1261.
[2] D. Wei, H. Feng, Q. Han, et al., Fault detection and diagnosis for variable-air-volume systems using combined residual, qualitative and quantitative techniques, Energy Build. 254 (2022) 111491.
[3] A.H. Abdullah, S.K. Yaman, H. Mohammad, et al., Construction manager's technical competencies in Malaysian construction projects, Eng. Construct. Architect. Manag. 25 (2) (2018) 153–177.
[4] D. Arditi, S.M.R. Alavipour, Trends in expectations about duties and responsibilities of construction managers, J. Construct. Eng. Manag. 145 (7) (2019) 04019037.
[5] K. Liu, Q. Meng, Q. Kong, et al., Review on the developments of structure, construction automation, and monitoring of intelligent construction, Buildings 12 (11) (2022) 1890.
[6] L. Zhang, Y. Li, Y. Pan, et al., Advanced informatic technologies for intelligent construction: a review, Eng. Appl. Artif. Intell. 137 (2024) 109104.
[7] Z. Guo, L. Li, A conceptual framework for collaborative development of intelligent construction and building industrialization, Front. Environ. Sci. 10 (2022) 904518.
[8] N. Li, S. Jiang, M. Wu, et al., Deformation monitoring for Chinese traditional timber buildings using fiber Bragg grating sensors, Sensors 18 (6) (2018) 1968.
[9] G. Yang, J. Wu, Q. Hu, et al., Real-time monitoring of changes in building tilt angle using double square artificial marks, Indoor Built Environ. 30 (6) (2021) 777–786.
[10] J. Yu, X. Meng, B. Yan, et al., Global Navigation Satellite System-based positioning technology for structural health monitoring: a review, Struct. Control Health Monit. 27 (1) (2020) e2467.
[11] S.M.A. Qaidi, Y.Z. Dinkha, J.H. Haido, et al., Engineering properties of sustainable green concrete incorporating eco-friendly aggregate of crumb rubber: a review, J. Clean. Prod. 324 (2021) 129251.
[12] S.M.A. Qaidi, B.A. Tayeh, A.M. Zeyad, et al., Recycling of mine tailings for the geopolymers production: a systematic review, Case Stud. Constr. Mater. 16 (2022) e00933 [EB/OL].
[13] M. Guo, H. Qi, Y. Zhao, et al., Design and management of a spatial database for monitoring building comfort and safety, Buildings 13 (12) (2023) 2982.
[14] Y. Liu, F. Xiao, Intelligent monitoring system of residential environment based on cloud computing and internet of things, IEEE Access 9 (2021) 58378–58389.
[15] D. Oboe, D. Poloni, C. Sbarufatti, et al., Towards automatic crack size estimation with ifem for structural health monitoring, Sensors 23 (7) (2023) 3406.
[16] H. Park, B. Oh, Real-time structural health monitoring of a supertall building under construction based on visual modal identification strategy, Autom. ConStruct. 85 (2018) 273–289.
[17] M. Dziendzikowski, P. Kozera, K. Kowalczyk, et al., Structural health monitoring of chemical storage tanks with application of PZT sensors, Sensors 23 (19) (2023) 8252.
[18] B. Mora, J. Basurko, I. Sabahi, et al., Strain virtual sensing for structural health monitoring under variable loads, Sensors 23 (10) (2023) 4706.
[19] V. Dang, H. Pham, Vibration-based building health monitoring using spatio-temporal learning model, Eng. Appl. Artif. Intell. 126 (2023) 106858.
[20] M. Guo, M. Sun, M. Huang, et al., High-precision measurement of steel structure based on LiDAR and UAV, Opt Precis. Eng. 29 (5) (2021) 989–998.
[21] Chen Y, Xie Y, Liang H, et al. Light frequency-hopping LiDAR and its anti-interference performance investigation [J/OL]. Adv. Laser Optoelectron.: 1-11[2022-07-23].
[22] Fu L, Zheng J, He H, et al. Decoupling SLAM method based on UAV 3D lidar for large indoor scenes[J/OL]. Electron. Meas. Technol.: 1-9[2022-07-23].
[23] Z. Liu, F. Zhang, X. Hong, Low-cost retina-like robotic lidars based on incommensurable scanning, IEEE ASME Trans. Mechatron. 27 (1) (2021) 58–68.
[24] M. Guo, H. Qi, K. Guo, et al., Ground-based LiDAR measurement system integrating Bei Dou/GNSS positioning and 5G communication, Opt Precis. Eng. 31 (4) (2023) 450–458.
[25] M. Guo, B. Yan, T. Zhou, et al., Application of LiDAR technology in deformation analysis of yingxian wooden pagoda, J. Architect. Civ. Eng. 37 (2) (2020) 109–117.

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